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Social Media Use Trajectories and Cognitive Performance in Adolescents

Social media is an increasingly important aspect of early adolescent lives, and trajectories of use have been correlated with negative psychosocial outcomes.^{1,2} Studies of the link between total screen time and cognitive performance in those aged 9 to 11 years from the Adolescent Brain Cognitive Development (ABCD)



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Supplemental content

study have shown mixed results.^{3,4} However, unlike passive screen time, social media use typically involves interactive, personalized, and cognitively demanding activities, yet there is a paucity of studies that have analyzed the associations between distinct longitudinal social media usage patterns and multiple domains of cognitive functioning.

This study examined the relationship between longitudinal patterns of social media use and cognitive performance 2 years later in a diverse, national sample of early adolescents. The study hypothesis was that greater social media use would be associated with lower cognitive performance.

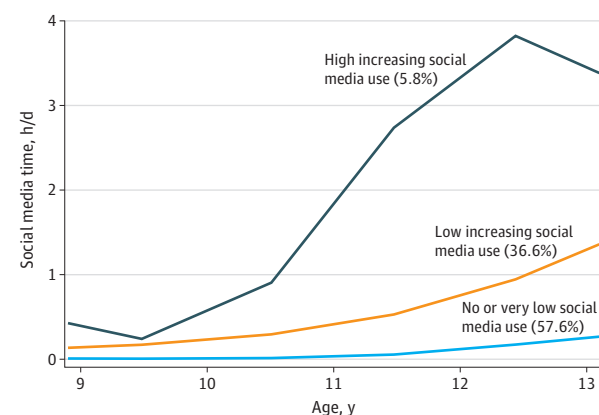
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Methods | Prospectively collected data from the ABCD study were analyzed from 3 time points: baseline (2016-2018, ages 9-10 years), year 1 (2017-2019), and year 2 (2018-2020). Institutional review board approval was obtained from the University of California, San Diego, and each study site. Written informed consent and assent were obtained from guardians and children, respectively.

Social media time (hours per day) trajectories from baseline to year 2 were determined using group-based trajectory modeling (eMethods in Supplement 1). This modeling characterized distinct patterns of social media use over time. Cognitive performance was determined using the National Institutes of Health (NIH) Toolbox Cognition Battery. Multiple linear regression analyses with robust standard errors estimated the association between social media trajectories and year 2 cognitive performance scores (5 subtests and composite). Models adjusted for baseline variables including age, sex, race and ethnicity, household income, parent education, attention-deficit/hyperactivity symptoms, depression symptoms, respective baseline cognitive score, other screen time, and study site. Analyses were weighted to approximate the American Community Survey by the US Census.

Results | In the analytic sample of 6554 adolescents, 51.1% were male and 48.9% were female. Three trajectory groups emerged over ages 9 to 13 years: no or very low social media

Figure. Social Media Time Trajectories by Age



use (57.6%, +0.3 h/d by age 13 years), low increasing social media use (36.6%, +1.3 h/d), and high increasing social media use (5.8%, +3.0 h/d) (Figure). The following findings are comparisons with the reference group, no or very low social media use. Low increasing social media use was associated with lower performance scores (negative differences in mean) on the Oral Reading Recognition Test (ORRT) (-1.39 [95% CI, -2.10 to -0.68]), Picture Sequence Memory Test (PSMT) (-2.03 [95% CI, -2.91 to -1.15]), Picture Vocabulary Test (PVT) (-2.09 [95% CI, -2.78 to -1.41]), and total composite score (-0.85 [95% CI, -1.30 to -0.40]) (Table). High increasing social media use was associated with poorer performance on the ORRT (-1.68 [95% CI, -3.08 to -0.28]), PSMT (-4.51 [95% CI, -6.48 to -2.54]), PVT (-3.85 [95% CI, -5.34 to -2.37]), and total composite score (-1.76 [95% CI, -2.72 to -0.81]).

Discussion | This analysis found that both low and high increases in social media use throughout early adolescence were significantly associated with lower performance in specific aspects of cognitive function, supporting a prior finding that greater screen time was negatively but weakly associated with adolescent cognitive performance.⁵ Overall, the differences in mean scores were small relative to the NIH Toolbox standard deviation of 15. Previous literature has hypothesized that social media use replacing more educational activities or schoolwork may explain the association between social media use and lower cognitive performance.⁶ The specific associations between increasing social media use and poorer performance on the ORRT and PVT, which tests stored language knowledge, support this hypothesis. The finding that even low levels of early adolescent social media exposure were linked to poorer cognitive performance may suggest support for stricter age restrictions. Limitations of this study include self-reported social media data, potential residual confounding, and an observational study design that precludes inferring that social media trajectories cause worse cognitive performance. Future studies should examine how specific social media platforms and content relate to cognitive outcomes.

Table. Cognitive Functioning Scores Across Social Media Trajectories (Ages 9-13 Years) in the Adolescent Brain Cognitive Development Study (N = 6554)^a

	Social media time trajectory				
	No or very low use, unadjusted mean (SD)	Low increasing, unadjusted mean (SD)	High increasing, unadjusted mean (SD)		
Flanker test	97.6 (14.2)	95.4 (14.0)	94.3 (14.6)		
Oral reading recognition	103.5 (16.9)	99.4 (15.9)	96.7 (14.1)		
Picture sequence memory	108.2 (16.2)	104.7 (16.5)	100.3 (16.1)		
Pattern comparison processing speed	108.3 (20.8)	107.8 (20.7)	106.9 (21.3)		
Picture vocabulary	106.3 (15.6)	99.6 (14.6)	94.8 (14.4)		
Total composite score	104.8 (10.6)	101.4 (10.4)	98.6 (10.4)		
	Adjusted mean (95% CI)	Adjusted mean (95% CI)	Adjusted mean (95% CI)		
Flanker test	96.9 (96.4 to 97.4)	96.2 (95.7 to 96.8)	96.0 (94.3 to 97.7)		
Oral reading recognition	102.2 (101.8 to 102.7)	100.8 (100.3 to 101.3)	100.5 (99.3 to 101.8)		
Picture sequence memory	107.5 (106.9 to 108.0)	105.4 (104.8 to 106.1)	103.0 (101.1 to 104.8)		
Pattern comparison processing speed	107.7 (107.0 to 108.4)	108.4 (107.5 to 109.2)	109.3 (107.1 to 111.6)		
Picture vocabulary	104.2 (103.8 to 104.6)	102.1 (101.6 to 102.6)	100.3 (98.9 to 101.7)		
Total composite score	103.6 (103.3 to 103.9)	102.8 (102.4 to 103.1)	101.8 (101.0 to 102.7)		
	Difference in mean (95% CI)	Difference in mean (95% CI)	P value	Difference in mean (95% CI)	P value
Flanker test	[Reference]	-0.67 (-1.45 to 0.11)	.09	-0.90 (-2.72 to 0.92)	.33
Oral reading recognition	[Reference]	-1.39 (-2.10 to -0.68)	<.001	-1.68 (-3.08 to -0.28)	.02
Picture sequence memory	[Reference]	-2.03 (-2.91 to -1.15)	<.001	-4.51 (-6.48 to -2.54)	<.001
Pattern comparison processing speed	[Reference]	0.65 (-0.48 to 1.78)	.26	1.61 (-0.75 to 3.98)	.18
Picture vocabulary	[Reference]	-2.09 (-2.78 to -1.41)	<.001	-3.85 (-5.34 to -2.37)	<.001
Total composite score	[Reference]	-0.85 (-1.30 to -0.40)	<.001	-1.76 (-2.72 to -0.81)	<.001

^a Sampling weights based on the American Community Survey were used to represent population estimates for unadjusted and adjusted means. Models represent the abbreviated output from the linear regression models with

adjustment for age, sex, race and ethnicity, household income, parent education, attention-deficit/hyperactivity symptoms, depression symptoms, respective baseline cognitive score, other screen time, and site location.

Jason M. Nagata, MD, MSc
 Jennifer H. Wong, MPH
 Kristen E. Kim, BA
 Racquel A. Richardson, BA
 Sahana Nayak
 Char Potes, BA
 Andreas M. Rauschecker, MD, PhD
 Aaron Scheffler, PhD, MS
 Leo P. Sugrue, MD, PhD
 Fiona C. Baker, PhD
 Alexander Testa, PhD

Author Affiliations: Division of Adolescent and Young Adult Medicine, Department of Pediatrics, University of California, San Francisco (Nagata, Wong, Kim, Richardson, Nayak, Potes); Department of Radiology and Biomedical Imaging, University of California, San Francisco (Rauschecker, Sugrue); Department of Epidemiology and Biostatistics, University of California, San Francisco (Scheffler); Department of Psychiatry and Behavioral Sciences, University of California, San Francisco (Sugrue); Center for Health Sciences, SRI International, Menlo Park, California (Baker); Department of Management, Policy and Community Health, University of Texas Health Science Center at Houston (Testa).

Accepted for Publication: August 20, 2025.

Published Online: October 13, 2025. doi:10.1001/jama.2025.16613

Corresponding Author: Jason M. Nagata, MD, MSc, 550 16th St, Fourth Floor, Box 0503, San Francisco, CA 94143 (jason.nagata@ucsf.edu).

Author Contributions: Ms Wong and Ms Kim had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.

Concept and design: Nagata, Wong, Richardson, Nayak, Potes, Testa.

Acquisition, analysis, or interpretation of data: Nagata, Wong, Kim, Potes, Rauschecker, Scheffler, Sugrue, Baker, Testa.

Drafting of the manuscript: Nagata, Wong, Kim, Richardson, Nayak, Potes, Sugrue.

Critical review of the manuscript for important intellectual content: Nagata, Wong, Richardson, Nayak, Potes, Rauschecker, Scheffler, Sugrue, Baker, Testa.

Statistical analysis: Nagata, Wong, Kim, Rauschecker, Scheffler, Sugrue, Testa.

Obtained funding: Nagata, Baker.

Administrative, technical, or material support: Nagata, Rauschecker, Testa.

Supervision: Nagata, Rauschecker, Testa.

Conflict of Interest Disclosures: Ms Kim reported receiving grants from Western Region Public Health Training Center outside the submitted work. Ms Richardson reported receiving grants from Western Region Public Health Training Center outside the submitted work. Mr Potes reported receiving grants from Western Region Public Health Training Center outside the submitted work. Dr Rauschecker reported receiving grants from GE Healthcare, American Society of Neuroradiology, National Comprehensive Cancer Center, and UCSF-UCB Schwab Dyslexia Center outside the submitted work. No other disclosures were reported.

Funding/Support: The research was supported by the National Institutes of Health (NIH) (K08HL159350, R01DA064134, and R01MH135492).

Role of the Funder/Sponsor: The NIH had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Data Sharing Statement: See Supplement 2.

Additional Information: The ABCD Study was supported by the NIH and additional federal partners under award numbers U01DA041022, U01DA041025, U01DA041028, U01DA041048, U01DA041089, U01DA041093, U01DA041106, U01DA041117, U01DA041120, U01DA041134, U01DA041148, U01DA041156, U01DA041174, U24DA041123, and U24DA041147. A full list of supporters is available at <https://abcdstudy.org/>

federal-partners.org/. A listing of participating sites and a complete listing of the study investigators can be found at <https://abcdstudy.org/principal-investigators.html>. ABCD consortium investigators designed and implemented the study and/or provided data but did not necessarily participate in analysis or writing of this report.

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COMMENT & RESPONSE

Intra-Arterial Tenecteplase After Successful Endovascular Therapy for Acute Stroke

To the Editor The ANGEL-TNK trial¹ provides valuable insights into the efficacy and safety of tenecteplase in acute ischemic stroke. However, we believe there are several methodological considerations that warrant further discussion. Specifically, we would like to address 3 issues: primary end point selection, statistical considerations, and control group design.

The trial's choice of an excellent outcome at 90 days, defined as a modified Rankin Scale (mRS) score of 0 or 1, as the sole primary end point is clinically meaningful but raises important questions. First, dichotomization of the mRS score at 0 to 1, while intuitive, may not capture the full spectrum of disability. The field increasingly recognizes the benefits of ordinal approaches, which offer greater statistical power and capture the full range of disability.² Dichotomization of the primary outcome variable may miss important treatment effects. Second, the significant primary outcome (40.5% with an excellent outcome in the intra-arterial tenecteplase group vs 26.4% in the standard medical treatment group; $P = .02$) contrasts with a nonsignificant ordinal shift ($P = .41$) and comparable rates of mRS scores of 0 to 2 (46.0% vs 46.5%) between the groups. This inconsistency suggests the primary result may represent a cherry-picked favorable outcome, rather than a robust treatment effect.

Several statistical issues also merit attention. First, with 7 secondary efficacy end points and 3 safety end points, it is unclear whether adjustments were made for multiple testing. The lone significant primary outcome ($P = .02$) could represent a type I error (rejection of a true null hypothesis). Second, the reported subgroup interactions (eg, in groups with and without glucose ≥ 100 mg/dL, internal carotid artery occlusions) appear underpowered and lack clear biological plausibility. It is also unclear whether these analyses were prespecified or exploratory.

Additionally, the standard medical treatment group raises several concerns. Unlike the CHOICE trial,³ which used a placebo-controlled design, the ANGEL-TNK trial did not account for procedural biases such as catheter placement effects (ie, it is hard to separate whether improvements are due to the treatment or just the act of performing the catheter procedure), contrast administration variables (eg, contrast media type and iodine concentration), and operator time differences.

While the ANGEL-TNK trial provides important safety data, these methodological considerations suggest a cautious interpretation of the efficacy results. We agree with the authors that confirmatory trials are needed, ideally incorporating these design improvements.

Yun Tao Zhao, MD, PhD

Author Affiliation: Aerospace Center Hospital, Beijing, China.

Corresponding Author: Yun Tao Zhao, MD, PhD, Aerospace Center Hospital, 15 Yuquan Rd, Haidian District, Beijing 100049, China (raas@hsc.pku.edu.cn).

Published Online: October 23, 2025. doi:10.1001/jama.2025.16270

Conflict of Interest Disclosures: None reported.

Disclaimer: The views expressed herein are those of the authors and do not reflect the official policy or position of the Veterans Affairs Administration or the People's Republic of China government or any of its affiliated entities.

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In Reply We thank Dr Zhao for pointing out these important issues about the trial. As we clarified in the Discussion section of the article,¹ the prespecified primary outcome of a 90-day mRS score of 0 to 1 was chosen because it was assumed that lytic benefit added to the already substantial benefit of endovascular therapy would cluster toward very favorable outcomes at the mRS score of 0 to 1, rather than mRS scores of 2 to 6 or the full disability spectrum with ordinal shift analysis.² We also acknowledged the inconsistent results for the ordinal shift and that using an mRS score of 0 to 2 might challenge the robustness of the treatment effect. As we clarified in the Methods section of the article,¹ we did not account for multiple comparisons of the secondary end points, as well as the subgroup analyses, and thus these findings should be interpreted as exploratory. The subgroup interactions in glucose level and internal carotid artery occlusions were underpowered, although these subgroup analyses were prespecified. Additionally, we acknowledged that as an open-label trial, the ANGEL-TNK trial did not account for procedural biases such as catheter placement effects, contrast administration variables, and operator time differences. However, we believe the effect due to these procedural biases might be minimal